

Achievement System User Documentation

Purpose:

The `AchievementManager` script handles the triggering and display of achievements within a Unity game. It provides a simple and extensible way to notify players when they unlock achievements.

Usage:

- Create Achievement Scriptable Objects:**
 - Create scriptable objects (SOs) that inherit from `AchievementSO`. These SOs will hold the data for each achievement (e.g., title, description, icon).
- Customize Achievement Popups:**
 - Design and customize the appearance of achievement popups by modifying the `achievementCard` prefab. Ensure the prefab has an `AchievementCard` script attached to it, which will handle displaying the achievement data.
- Create an Achievement Manager GameObject:**
 - Create an empty GameObject in your scene and attach the `AchievementManager` script to it.
 - Assign the customized `achievementCard` prefab to the `achievementCard` field in the Inspector.
 - Assign the desired parent transform for the achievement cards to the `achievementTransform` field.
- Trigger Achievements:**
 - To trigger an achievement from any other script, simply invoke the `AchievementManager.triggerAchievement` event, passing the corresponding `AchievementSO` as an argument.
 - Example:

```
AchievementManager.triggerAchievement?.Invoke(myAchievementSO);
```
- AchievementCard Script:**
 - You are required to create a script named `AchievementCard` that is attached to the `achievementCard` prefab. This script should have a public method named `Initialize(AchievementSO achievementSO)` that will be called by the `AchievementManager` when creating the achievement popup. This script will handle the display of the data from the `AchievementSO` onto the popup.
- AchievementSO Scriptable Object:**
 - You are required to create a scriptable object that inherits from `AchievementSO`. This scriptable object should contain the data that will be displayed on the achievement popup. Example:

C#

```
using UnityEngine;
```

```
[CreateAssetMenu(fileName = "New Achievement", menuName =  
"Achievements/Achievement")]  
public class AchievementSO : ScriptableObject
```

```

{
    public string title;
    public string description;
    public Sprite icon;
}

```

7.

Key Components:

- **triggerAchievement (Delegate and Static Event):**
 - A static event that allows other scripts to trigger achievements.
 - Scripts can subscribe to this event to listen for achievement triggers.
- **achievementCard (GameObject):**
 - The prefab used to display achievement notifications.
- **achievementTransform (Transform):**
 - The parent transform under which achievement cards will be instantiated.
- **OnAchievementTriggered(AchievementSO achievementSO) (Method):**
 - Instantiates and initializes an achievement card when an achievement is triggered.
- **AchievementCard (Component):**
 - This component should be attached to the achievementCard prefab, and handle the display of the achievement data.
- **AchievementSO (Scriptable Object):**
 - This scriptable object should hold the data for each achievement.

Workflow:

1. An event (e.g., completing a level, collecting an item) triggers the `AchievementManager.triggerAchievement` event.
2. The `OnAchievementTriggered` method in `AchievementManager` is called.
3. An instance of the `achievementCard` prefab is created.
4. The `AchievementCard` component on the instantiated card is initialized with the achievement data from the `AchievementSO`.
5. The achievement card is displayed to the player.